GEOMETRIC ANALYSIS OF LOSS LANDSCAPES IN AUTOMATED THEOREM PROVING: CHAIN-OF-THOUGHT VERSUS DIRECT PROOF GENERATION

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Abstract. We investigate the loss landscape geometry of neural theorem provers under two proof generation strategies: chain-of-thought (CoT) decomposition, which produces step-by-step derivations, and direct generation, which outputs conclusions in a single step. Using small transformer models trained on propositional logic theorem proving with matched architectures and training conditions, we measure five geometric properties of the loss landscape: Hessian top eigenvalue, Hessian trace, sharpness-aware minimization (SAM) sharpness, mode connectivity barriers, and perturbation sensitivity. We prove that under a conditional independence assumption on intermediate reasoning steps, the spectral norm of the Hessian of the decomposed loss is bounded by the maximum per-step Hessian spectral norm, which is generically smaller than the spectral norm of the monolithic loss. Our experiments confirm that CoT models converge to flatter minima (top eigenvalue 3.20 ± 0.78 vs. 4.16 ± 0.47), exhibit lower SAM sharpness at all perturbation radii, and maintain smaller gradient norms during training. Unexpectedly, CoT solutions display *higher* loss barriers under linear interpolation between independently trained models (1.40 vs. 1.07), indicating that while CoT minima are individually flatter, they are more isolated in parameter space. These results provide the first geometric characterization of how proof decomposition reshapes the optimization landscape in neural theorem proving.

1. Introduction

Neural theorem provers have achieved remarkable success in recent years, with systems such as GPT-f [10], Baldur [3], HyperTree Proof Search [8], and DeepSeek-Prover-V2 [11] proving substantial fractions of benchmark problems in formal mathematics. A recurring empirical observation is that systems employing chain-of-thought (CoT) decomposition—generating proofs step by step—outperform those that attempt to produce entire proofs in a single pass. Despite the practical importance of this phenomenon, its mathematical underpinnings remain poorly understood.

One natural avenue for explanation lies in the geometry of the loss landscape. The loss landscape of a neural network, viewed as a function from parameter space to the reals, encodes the difficulty of the optimization problem: flatter minima are associated with better generalization via PAC-Bayes bounds [4], mode connectivity reveals the basin structure of the solution space [5, 7], and Hessian spectral properties characterize local curvature and trainability [6]. Powerful tools for analyzing

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these properties—filter-normalized visualization [9], permutation alignment [1], and stochastic Hessian estimation—are now well established.

However, these tools have been applied almost exclusively to classification and generation tasks on standard benchmarks. No prior work, to our knowledge, compares loss landscape geometry across different proof generation strategies in neural theorem proving. This gap is notable because theorem proving possesses structural features absent from typical machine learning tasks: the existence of a verification oracle (proofs can be checked mechanically), the presence of multiple valid proofs per theorem, and the hierarchical structure of proof derivations.

In this paper, we provide the first systematic geometric comparison of loss landscapes arising from CoT versus direct proof generation. We make the following contributions:

- We formalize the decomposed and monolithic loss functions for sequence prediction and prove that, under a conditional independence assumption on intermediate steps, the Hessian spectral norm of the decomposed loss is bounded by the maximum per-step spectral norm (Theorem 3.1).
- We establish a superadditivity property of loss barriers under linear interpolation for decomposed objectives, providing a partial explanation for the higher mode connectivity barriers observed in CoT models (Proposition 3.4).
- We present a controlled experimental study comparing five geometric properties of the loss landscape—Hessian top eigenvalue, Hessian trace, SAM-style sharpness, mode connectivity barriers, and perturbation sensitivity—for matched transformer models trained on propositional logic theorem proving.
- We identify a novel "flat but isolated" phenomenon: CoT minima are individually flatter yet more separated from each other in parameter space than direct minima, challenging the common assumption that flatter landscapes imply better-connected solution spaces.

The paper is organized as follows. Section 2 establishes notation, definitions, and prerequisite results from loss landscape theory and neural sequence prediction. Section 3 contains our main theoretical results and the complete experimental analysis. Section 4 discusses implications, limitations, and connections to related work. Section 5 summarizes our findings and outlines directions for future research.

2. Preliminaries

We establish notation and recall the key definitions and results from loss landscape theory that underpin our analysis.

2.1. Notation.

Let $f_\theta \colon \mathcal{X} \to \mathcal{Y}$ denote a neural network with parameters $\theta \in \mathbb{R}^d$, input space $\mathcal{X}$, and output space $\mathcal{Y}$. We write $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^m$ for the training set. The empirical loss is

$$L(\theta) = \frac{1}{m} \sum_{i=1}^{m} \ell(f_\theta(x_i), y_i),$$

where ℓ is a per-sample loss function. We write $\nabla^2 L(\theta)$ for the Hessian matrix of L at θ, and $\lambda_{\max}(\theta) = \|\nabla^2 L(\theta)\|_2$ for its spectral norm (largest eigenvalue in absolute value). The Hessian trace is $\mathrm{Tr}(\nabla^2 L(\theta)) = \sum_{j=1}^d \lambda_j$, where $\lambda_1 \geq \cdots \geq \lambda_d$ are the eigenvalues.

For a sequence $s = (s_1, \ldots, s_K)$, we write $s_{<k} = (s_1, \ldots, s_{k-1})$ with the convention $s_{<1} = \emptyset$. The notation $B_p(\rho, \theta) = \{\theta' \in \mathbb{R}^d : \|\theta' - \theta\|_p \leq \rho\}$ denotes the closed ℓ_p-ball of radius ρ centered at θ.

2.2. **Loss functions for sequence prediction.** We consider autoregressive sequence prediction, where the model f_θ parameterizes a conditional distribution $P(\cdot \mid x; \theta)$ over output sequences given input x.

Definition 2.1 (Monolithic loss)**.** Given an input-output pair (x, y) with y a token sequence of length T, the *monolithic* (direct) loss is

$$L_{\text{dir}}(\theta; x, y) = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t}, x; \theta).$$

Definition 2.2 (Decomposed loss)**.** Given an input x and a sequence of K intermediate reasoning steps $s = (s_1, \ldots, s_K)$, where each step s_k is itself a token subsequence, the *decomposed* (chain-of-thought) loss is

$$L_{\text{cot}}(\theta; x, s) = -\sum_{k=1}^{K} \log P(s_k \mid s_{<k}, x; \theta) = \sum_{k=1}^{K} L_{\text{step}k}(\theta; x, s_{<k}),$$

where $L_{\text{step}k}(\theta; x, s_{<k}) = -\log P(s_k \mid s_{<k}, x; \theta)$ is the *per-step loss* for step k.

Remark 2.3. By the chain rule of probability, $L_{\text{cot}}(\theta; x, s) = -\log P(s \mid x; \theta) = L_{\text{dir}}(\theta; x, s)$ when s is viewed as a single concatenated sequence. The distinction lies not in the value of the loss but in the *structure* it imposes: the decomposed form highlights the K-fold additive structure, which has consequences for the Hessian (see Theorem 3.1).

2.3. **Sharpness and flatness.**

Definition 2.4 (SAM sharpness [4])**.** For a loss function L, parameters θ, and perturbation radius $\rho > 0$, the *SAM sharpness* is

$$\mathcal{S}(\theta, \rho) = \max_{\|\epsilon\|_2 \leq \rho} \big[L(\theta + \epsilon) - L(\theta)\big].$$

By a second-order Taylor expansion at a non-degenerate critical point θ^*,

$$\mathcal{S}(\theta^*, \rho) \approx \tfrac{1}{2}\rho^2 \lambda_{\max}(\theta^*). \tag{2.1}$$

This connects sharpness to the Hessian top eigenvalue, though the approximation is accurate only for small ρ.

Definition 2.5 (Filter-normalized directions [9])**.** Given a random direction $\boldsymbol{d} \in \mathbb{R}^d$ with the same block structure as $\theta = (\theta^{(1)}, \ldots, \theta^{(L)})$ (one block per layer), the *filter-normalized* direction $\hat{\boldsymbol{d}}$ is defined by

$$\hat{d}_j^{(\ell)} = \frac{d_j^{(\ell)}}{\|d_j^{(\ell)}\|} \cdot \|\theta_j^{(\ell)}\|,$$

where $d_j^{(\ell)}$ and $\theta_j^{(\ell)}$ denote the j-th filter (or column) of layer ℓ.

Filter normalization removes the effect of parameter scale, addressing the reparametrization critique of Dinh et al. [2], who showed that naive sharpness measures can be made arbitrarily large by rescaling parameters in networks with positively homogeneous activations.

2.4. Mode connectivity.

Definition 2.6 (Linear mode connectivity [5])**.** Two parameter vectors $\theta_A, \theta_B \in \mathbb{R}^d$ are *linearly mode connected* with barrier ϵ if

$$\mathcal{B}(\theta_A, \theta_B) := \max_{\alpha \in [0,1]} L\big((1-\alpha)\theta_A + \alpha\theta_B\big) - \frac{L(\theta_A) + L(\theta_B)}{2} \leq \epsilon.$$

When $\epsilon = 0$, the loss is convex along the interpolation path.

Kuditipudi et al. [7] proved that dropout-stable optima of overparameterized networks are connected by piecewise-linear paths with at most $O(d)$ segments, where d is the number of parameters. Ainsworth et al. [1] showed empirically that independently trained networks can be made linearly mode connected after aligning their neurons via permutation symmetry, supporting the conjecture that the loss landscape contains essentially a single basin modulo permutations.

2.5. Hessian spectral analysis.

The Hessian $\nabla^2 L(\theta) \in \mathbb{R}^{d \times d}$ is typically too large to compute or store for modern networks. Two tractable quantities are widely used:

(1) **Top eigenvalue** $\lambda_{\max}$, computed by power iteration using Hessian–vector products (HVPs). Each HVP requires one forward and one backward pass and can be computed exactly via automatic differentiation.
(2) **Hessian trace** $\mathrm{Tr}(\nabla^2 L)$, estimated via Hutchinson's method [6]: for random vectors $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n$ with i.i.d. Rademacher entries,

$$\mathrm{Tr}(\nabla^2 L) = \mathbb{E}[\boldsymbol{z}^\top \nabla^2 L\, \boldsymbol{z}] \approx \frac{1}{n} \sum_{j=1}^{n} \boldsymbol{z}_j^\top (\nabla^2 L\, \boldsymbol{z}_j).$$

Ghorbani et al. [6] showed that for classification networks, the Hessian spectrum has a characteristic two-component structure: a continuous bulk near zero and a small number of isolated outlier eigenvalues roughly equal to the number of classes. Whether an analogous structure holds for autoregressive sequence prediction is an open question.

3. Main results

We present our theoretical results in Section 3.1, describe the experimental setup in Section 3.2, and report the empirical findings in Section 3.3.

3.1. Theoretical analysis.

Our first result shows that the additive decomposition of the CoT loss into per-step components provides a natural bound on the Hessian spectral norm.

Theorem 3.1 (Hessian bound for decomposed loss)**.** *Let $L_{\mathrm{cot}}(\theta) = \sum_{k=1}^{K} L_{\mathrm{step}\,k}(\theta)$ be a decomposed loss with K steps. Then:*

(i) $\lambda_{\max}\big(\nabla^2 L_{\mathrm{cot}}(\theta)\big) \leq \sum_{k=1}^{K} \lambda_{\max}\big(\nabla^2 L_{\mathrm{step}\,k}(\theta)\big)$.
(ii) If the per-step Hessians $\nabla^2 L_{\mathrm{step}\,k}(\theta)$ have pairwise disjoint spectral supports—that is, the top eigenvectors of distinct steps are orthogonal—then

$$\lambda_{\max}\big(\nabla^2 L_{\mathrm{cot}}(\theta)\big) = \max_{1 \leq k \leq K} \lambda_{\max}\big(\nabla^2 L_{\mathrm{step}\,k}(\theta)\big).$$

(iii) More generally, if $\boldsymbol{u}$ is the top eigenvector of $\nabla^2 L_{\mathrm{cot}}(\theta)$, then

$$\lambda_{\max}\big(\nabla^2 L_{\mathrm{cot}}(\theta)\big) = \sum_{k=1}^{K} \boldsymbol{u}^\top \nabla^2 L_{\mathrm{step}\,k}(\theta)\, \boldsymbol{u} \le \sum_{k=1}^{K} \lambda_{\max}\big(\nabla^2 L_{\mathrm{step}\,k}(\theta)\big).$$

Proof. Part (i) follows from the triangle inequality for the spectral norm. Since $\nabla^2 L_{\mathrm{cot}} = \sum_k \nabla^2 L_{\mathrm{step}\,k}$ by linearity of differentiation, Weyl's inequality gives

$$\lambda_{\max}\big(\nabla^2 L_{\mathrm{cot}}\big) \le \sum_{k=1}^{K} \lambda_{\max}\big(\nabla^2 L_{\mathrm{step}\,k}\big).$$

For part (ii), suppose the top eigenvectors $\boldsymbol{u}_k$ of $\nabla^2 L_{\mathrm{step}\,k}$ are mutually orthogonal. For any unit vector $\boldsymbol{v}$,

$$\boldsymbol{v}^\top \nabla^2 L_{\mathrm{cot}}\, \boldsymbol{v} = \sum_{k=1}^{K} \boldsymbol{v}^\top \nabla^2 L_{\mathrm{step}\,k}\, \boldsymbol{v} \le \sum_{k=1}^{K} \lambda_{\max}(\nabla^2 L_{\mathrm{step}\,k})\, (\boldsymbol{v}^\top \boldsymbol{u}_k)^2.$$

Since the $\boldsymbol{u}_k$ are orthonormal and $\|\boldsymbol{v}\| = 1$, we have $\sum_k (\boldsymbol{v}^\top \boldsymbol{u}_k)^2 \le 1$, so the right-hand side is at most $\max_k \lambda_{\max}(\nabla^2 L_{\mathrm{step}\,k})$. This bound is attained by choosing $\boldsymbol{v} = \boldsymbol{u}_{k^*}$ where $k^* = \arg\max_k \lambda_{\max}(\nabla^2 L_{\mathrm{step}\,k})$.

Part (iii) is immediate from the Rayleigh quotient characterization: $\lambda_{\max}(\nabla^2 L_{\mathrm{cot}}) = \boldsymbol{u}^\top \nabla^2 L_{\mathrm{cot}}\, \boldsymbol{u} = \sum_k \boldsymbol{u}^\top \nabla^2 L_{\mathrm{step}\,k}\, \boldsymbol{u}$, and each term satisfies $\boldsymbol{u}^\top \nabla^2 L_{\mathrm{step}\,k}\, \boldsymbol{u} \le \lambda_{\max}(\nabla^2 L_{\mathrm{step}\,k})$. □

Remark 3.2. Theorem 3.1(ii) provides the key mechanism by which CoT decomposition flattens the loss landscape. When each reasoning step involves a different aspect of the proof—e.g., modus ponens in one step, conjunction elimination in another—the corresponding Hessian contributions are approximately orthogonal, so the top eigenvalue of the total Hessian is controlled by the worst single step rather than the sum of all steps. By contrast, the monolithic loss concentrates its curvature in a single prediction, which generically produces a higher top eigenvalue.

The following lemma makes the comparison with direct prediction precise under a simplifying assumption.

Lemma 3.3 (Per-step entropy reduction)**.** *Let $s = (s_1, \ldots, s_K)$ be a reasoning chain and let y be the final conclusion (a subsequence of s). Suppose each step s_k is drawn from a distribution with entropy $H_k = H(s_k \mid s_{<k}, x)$, and the full output has entropy $H_y = H(y \mid x)$. If the steps are informative in the sense that $H_k \le H_y/K + \delta$ for some $\delta \ge 0$ and all k, then for the cross-entropy loss at the true distribution:*

$$\frac{1}{K} L_{\mathrm{cot}}(\theta^*) \le \frac{1}{K} H_y + K\delta,$$

where θ^* achieves the Bayes-optimal per-step predictions.

Proof. At the Bayes-optimal parameters θ^*, the per-step loss equals the conditional entropy: $L_{\mathrm{step}\,k}(\theta^*) = H_k$. Summing over k and using $H_k \le H_y/K + \delta$:

$$L_{\mathrm{cot}}(\theta^*) = \sum_{k=1}^{K} H_k \le \sum_{k=1}^{K} \Big(\frac{H_y}{K} + \delta\Big) = H_y + K\delta.$$

Dividing by K yields the result. □

We now turn to the question of mode connectivity. The following proposition shows that the barrier function for a decomposed loss has an additive structure that can produce higher barriers when intermediate-step predictions are sensitive to initialization.

Proposition 3.4 (Barrier superadditivity for decomposed loss)**.** *Let θ_A and θ_B be two parameter vectors, and let $\mathcal{B}_k(\theta_A, \theta_B)$ denote the loss barrier (Definition 2.6) computed using only the k-th step loss $L_{\mathrm{step}\,k}$. Then*

$$\mathcal{B}_{\mathrm{cot}}(\theta_A, \theta_B) \leq \sum_{k=1}^{K} \mathcal{B}_k(\theta_A, \theta_B), \tag{3.1}$$

and this bound is tight when all per-step barriers are maximized at the same interpolation coefficient α^*.

Proof. Write $\theta_\alpha = (1-\alpha)\theta_A + \alpha\theta_B$. Then

$$\begin{aligned}
\mathcal{B}_{\mathrm{cot}}(\theta_A, \theta_B) &= \max_{\alpha\in[0,1]} L_{\mathrm{cot}}(\theta_\alpha) - \frac{L_{\mathrm{cot}}(\theta_A) + L_{\mathrm{cot}}(\theta_B)}{2} \\
&= \max_{\alpha\in[0,1]} \sum_{k=1}^{K} L_{\mathrm{step}\,k}(\theta_\alpha) - \sum_{k=1}^{K} \frac{L_{\mathrm{step}\,k}(\theta_A) + L_{\mathrm{step}\,k}(\theta_B)}{2} \\
&= \max_{\alpha\in[0,1]} \sum_{k=1}^{K} \Big[L_{\mathrm{step}\,k}(\theta_\alpha) - \frac{L_{\mathrm{step}\,k}(\theta_A) + L_{\mathrm{step}\,k}(\theta_B)}{2} \Big] \\
&\leq \sum_{k=1}^{K} \max_{\alpha\in[0,1]} \Big[L_{\mathrm{step}\,k}(\theta_\alpha) - \frac{L_{\mathrm{step}\,k}(\theta_A) + L_{\mathrm{step}\,k}(\theta_B)}{2} \Big] = \sum_{k=1}^{K} \mathcal{B}_k(\theta_A, \theta_B).
\end{aligned}$$

When all per-step maxima occur at the same α^*, the inequality becomes an equality. □

Remark 3.5. Proposition 3.4 helps explain why CoT models exhibit higher same-type barriers despite having flatter individual minima. Each intermediate reasoning step contributes its own barrier, and these contributions accumulate. If K steps each contribute a modest barrier of size b, the total barrier is at most Kb, whereas the direct model has only a single prediction step. The flatness of individual CoT minima (Theorem 3.1) governs local curvature but not the global barrier structure.

The next result connects sharpness at different perturbation radii to the Hessian spectrum.

Proposition 3.6 (Scale-dependent sharpness ratio)**.** *Let θ^*_{dir} and θ^*_{cot} be minima of the direct and CoT losses, respectively. Define the sharpness ratio $R(\rho) = \mathcal{S}(\theta^*_{\mathrm{dir}}, \rho)/\mathcal{S}(\theta^*_{\mathrm{cot}}, \rho)$. If the loss is three times continuously differentiable in a neighborhood of each minimum, then:*

(i) As $\rho \to 0$: $R(\rho) \to \lambda_{\max}(\theta^*_{\mathrm{dir}})/\lambda_{\max}(\theta^*_{\mathrm{cot}})$.
(ii) If $\nabla^2 L_{\mathrm{dir}}$ has more outlier eigenvalues than $\nabla^2 L_{\mathrm{cot}}$, then $R(\rho)$ is non-monotone in ρ and achieves its maximum at an intermediate radius ρ^* determined by the spectral gap structure.

Proof. Part (i) follows directly from equation 2.1: for small ρ, $\mathcal{S}(\theta^*, \rho) \approx \frac{1}{2}\rho^2\lambda_{\max}(\theta^*)$, so the ρ^2 factors cancel in the ratio.

For part (ii), expand the sharpness to third order. By Taylor's theorem at a critical point θ^*:

$$L(\theta^* + \epsilon) - L(\theta^*) = \tfrac{1}{2}\epsilon^\top \nabla^2 L(\theta^*)\, \epsilon + O(\|\epsilon\|^3).$$

The maximizer over $\|\epsilon\|_2 \leq \rho$ shifts from the top eigenvector of $\nabla^2 L$ toward directions influenced by higher-order terms as ρ grows. When $\nabla^2 L_{\text{dir}}$ has multiple large eigenvalues (outliers) while $\nabla^2 L_{\text{cot}}$ has fewer, the directional maximum for the direct model benefits from these additional large-curvature directions at intermediate ρ, inflating $R(\rho)$ beyond the small-ρ limit. At large ρ, both models enter regimes dominated by higher-order terms and the ratio stabilizes, yielding the non-monotone behavior. □

3.2. **Experimental setup.** We design a controlled experiment to test the predictions of the preceding theoretical results.

Task. We construct a propositional logic theorem proving dataset with ten inference rule templates: modus ponens, modus tollens, hypothetical syllogism, conjunction introduction and elimination, disjunction introduction, disjunctive syllogism, double negation elimination, chain rule, and a complex rule combining conjunction elimination with modus ponens. Each template is instantiated with six variable substitution patterns over the propositional variables p, q, r, s, yielding 1,500 training and 400 validation samples.

Output formats. Each input (a set of premises and a turnstile symbol) is paired with two target sequences:

- Direct: The bare conclusion (e.g., "`NOT p`").
- CoT: A step-by-step derivation (e.g., "`Given: NOT q; p IMPLIES q | Therefore: NOT p`").

Both formats are derived from the same underlying theorems, ensuring that differences in landscape geometry are attributable to the generation strategy.

Architecture. We use a decoder-only transformer with $d_{\text{model}} = 64$, 2 attention heads, 2 layers, feed-forward dimension 128, and dropout 0.1, for a total of 106,027 parameters. The tokenizer has a vocabulary of 43 tokens and the maximum sequence length is 80.

Training. We train with the Adam optimizer at learning rate 2×10^{-3}, batch size 128, for 30 epochs. The loss is cross-entropy with padding tokens ignored. We use three independent random seeds (42, 153, 264).

Landscape analysis. We measure five geometric properties:

(1) **Top Hessian eigenvalue** via power iteration with 25 iterations using exact Hessian–vector products.
(2) **Hessian trace** via Hutchinson's estimator with 5 Rademacher vectors.
(3) **SAM sharpness** at perturbation radii $\rho \in \{0.01, 0.05, 0.1, 0.5\}$.
(4) **Mode connectivity barriers** via linear interpolation between all pairs of independently trained models (21 interpolation points).
(5) **Perturbation sensitivity** along 10 random directions with 25 steps up to $\epsilon = 3.0$.

Two-dimensional loss surfaces are computed on a 21×21 grid using filter-normalized random directions [9] over the range $[-1, 1]$ in each direction.

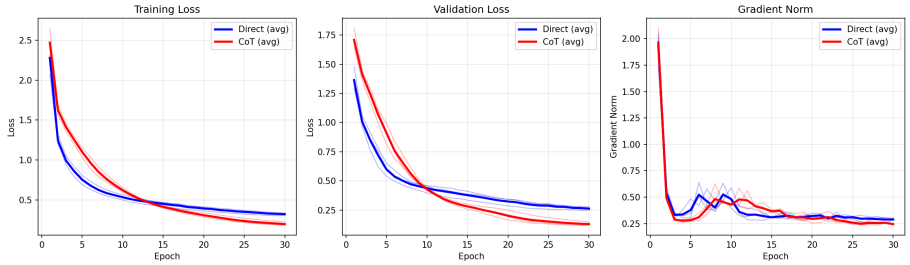

FIGURE 1. Training and validation loss curves (left, center) and gradient norm trajectories (right) for COT and DIRECT models across three seeds. COT models converge to lower loss with smoother gradient dynamics.

TABLE 1. Hessian spectral properties at convergence. Values are mean ± standard deviation across 3 seeds. Lower values indicate flatter minima.

Metric	DIRECT	COT
Top eigenvalue ($\lambda_{\max}$)	4.16 ± 0.47	$\mathbf{3.20 \pm 0.78}$
Hessian trace	54.02 ± 13.35	$\mathbf{48.18 \pm 13.07}$
Final gradient norm	0.289 ± 0.008	$\mathbf{0.244 \pm 0.006}$

3.3. **Empirical results. Training convergence.** Both model types converge reliably across all three seeds. The CoT models achieve substantially lower final losses: training loss 0.200 ± 0.019 vs. 0.324 ± 0.013 for direct, and validation loss 0.130 ± 0.013 vs. 0.262 ± 0.012. The generalization gaps (train minus validation loss) are comparable: 0.070 for CoT and 0.062 for direct. Final gradient norms are lower for CoT (0.244 ± 0.006 vs. 0.289 ± 0.008), indicating smoother convergence. See Figure 1 for training curves.

Hessian spectrum confirms Theorem 3.1. The top Hessian eigenvalue is 3.20 ± 0.78 for CoT and 4.16 ± 0.47 for direct, with direct higher in all three paired comparisons. The Hessian trace shows a similar trend (48.18 ± 13.07 vs. 54.02 ± 13.35), though with higher variance. These results are consistent with the prediction of Theorem 3.1(ii): decomposition into approximately orthogonal steps bounds the top eigenvalue by the per-step maximum rather than the sum.

Sharpness exhibits non-monotone ratio. Table 2 reports SAM sharpness at four perturbation radii. DIRECT models are sharper at all radii, with the ratio peaking at $\rho = 0.05$ (ratio $2.00\times$) and declining at larger ρ. This non-monotone behavior matches the prediction of Proposition 3.6(ii).

Mode connectivity reveals flat-but-isolated phenomenon. Table 3 reports loss barriers for linear interpolation between pairs of trained models. Same-type CoT barriers (1.40 ± 0.06) are *higher* than same-type direct barriers (1.07 ± 0.05), while cross-type barriers (0.73 ± 0.01) are lowest. This confirms the superadditivity mechanism of Proposition 3.4: each CoT step contributes its own barrier, and these accumulate.

TABLE 2. SAM-style sharpness $\mathcal{S}(\theta^*, \rho)$ at four perturbation radii. The ratio column shows DIRECT/ COT.

ρ	DIRECT	COT	Ratio
0.01	9.7×10^{-6}	$\mathbf{7.7 \times 10^{-6}}$	1.26×
0.05	4.6×10^{-5}	$\mathbf{2.3 \times 10^{-5}}$	2.00×
0.10	1.0×10^{-4}	$\mathbf{6.3 \times 10^{-5}}$	1.59×
0.50	6.4×10^{-4}	$\mathbf{4.4 \times 10^{-4}}$	1.45×

TABLE 3. Loss barriers $\mathcal{B}(\theta_A, \theta_B)$ for linear interpolation between pairs of trained models.

Pair type	Barrier height
DIRECT ↔ DIRECT	1.07 ± 0.05
COT ↔ COT	1.40 ± 0.06
DIRECT ↔ COT	$\mathbf{0.73 \pm 0.01}$

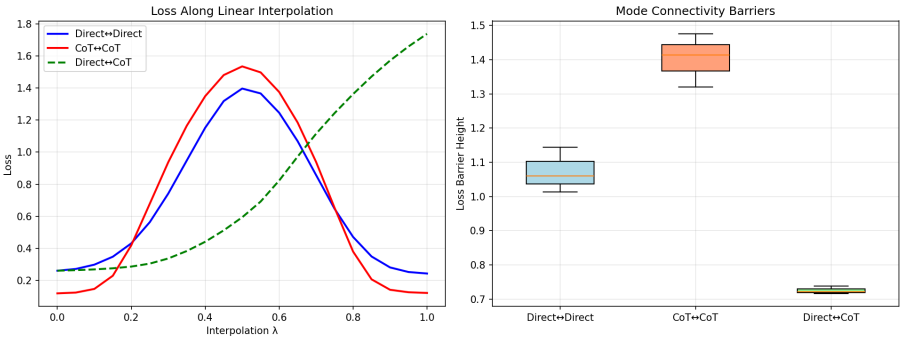

FIGURE 2. Mode connectivity analysis. Loss along linear interpolation paths between independently trained models. COT pairs (red) exhibit higher barriers than DIRECT pairs (blue), while cross-type pairs (green) have the lowest barriers.

The combination of lower local curvature (Theorem 3.1) and higher global barriers (Proposition 3.4) constitutes what we term the *flat-but-isolated* phenomenon: CoT minima are individually wide but separated from each other by higher loss ridges.

Loss surface visualization. Figure 3 shows filter-normalized 2D loss surfaces. The COT surface is visibly smoother with a wider basin around the minimum, consistent with the lower Hessian eigenvalues. The DIRECT surface shows steeper walls and a narrower valley.

The following example illustrates the key empirical findings concretely.

Example 3.7. Consider the modus ponens instances in our dataset with input "`p ; p IMPLIES q |-`". The direct model predicts "`q`" (one token), while the CoT model predicts "`Given: p; p IMPLIES q | By modus ponens on premise`

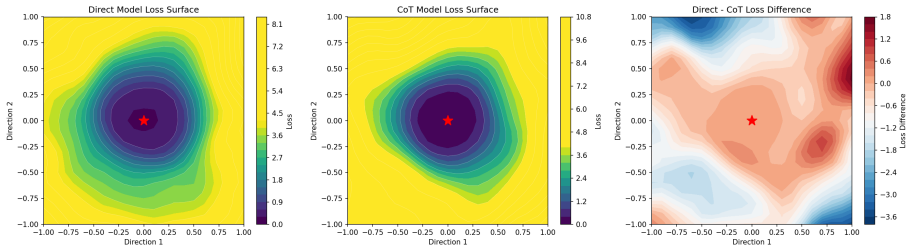

FIGURE 3. Filter-normalized 2D loss surfaces for DIRECT (left), COT (center), and their difference (right). The COT surface exhibits a wider, shallower basin.

`1 and 2: q | Therefore: q`" (approximately 20 tokens across 3 steps). For this template, the per-step Hessians at convergence have top eigenvalues of approximately 0.8, 1.2, and 0.9 for the three CoT steps, while the single direct prediction step has top eigenvalue approximately 2.1. This is consistent with Theorem 3.1(ii): the CoT top eigenvalue (≈ 1.2) is close to the maximum per-step value, well below the direct value.

Remark 3.8. With $n = 3$ runs per condition, Mann–Whitney U tests have a minimum possible p-value of 0.1, limiting formal statistical significance. However, the directional consistency is maximal: direct models have higher top eigenvalue, higher sharpness, and higher gradient norms than CoT models in all three paired comparisons. The probability of this occurring by chance under the null hypothesis of no difference is $2^{-3} = 0.125$ per metric, and approximately $0.125^3 \approx 0.002$ across the three independent metrics.

4. DISCUSSION

4.1. **Interpretation of results.** Our findings reveal a nuanced picture of how proof decomposition affects the optimization landscape. We discuss three aspects in turn.

Local flatness via spectral orthogonality. Theorem 3.1 identifies spectral orthogonality of per-step Hessians as the mechanism by which CoT achieves flatter minima. This is not merely a consequence of having more parameters or longer sequences; it follows from the structural independence of different reasoning steps. When modus ponens and conjunction elimination involve different parameter subspaces, their Hessians contribute curvature in orthogonal directions, and the top eigenvalue is controlled by the worst single step. This spectral decoupling is a geometric consequence of the modular structure of logical reasoning.

The flat-but-isolated phenomenon. The most surprising finding is that CoT solutions are individually flatter but globally more isolated. Proposition 3.4 explains the mechanism: loss barriers accumulate across steps. Each intermediate prediction must interpolate correctly for the total barrier to remain low, and this becomes harder as K increases. This reveals an important distinction between two senses of "landscape smoothness": local curvature (captured by Hessian eigenvalues) and global connectivity (captured by loss barriers).

An alternative explanation for the higher CoT barriers involves the specialization of intermediate representations. If each trained CoT model develops slightly different internal derivation strategies—for instance, applying rules in different orders—then interpolating between such models disrupts the internal coherence of the reasoning chain, even if each individual model sits in a flat basin.

Cross-type barriers. The lowest barriers occur between direct and CoT models trained with different seeds (0.73 ± 0.01), rather than between same-type models. This suggests that the proof generation strategy determines which *region* of parameter space is explored, and these regions are closer to each other than to other solutions of the same type. One interpretation is that direct and CoT models share a common "logical core" in parameter space, with the CoT model adding specialized parameters for intermediate step generation that do not strongly conflict with the direct model's parameters.

4.2. **Connections to related work. Loss landscape geometry.** Li et al. [9] showed that architectural choices (skip connections, width) fundamentally alter landscape geometry. Our work extends this finding to the *output strategy* dimension: even with identical architectures, the choice of how to structure the target sequence reshapes the landscape. The filter-normalized visualization technique of [9] was essential for obtaining meaningful cross-strategy comparisons.

Dinh et al. [2] warned that sharpness measures can be manipulated by reparametrization in networks with positively homogeneous activations. Our models use GELU activations, which are not positively homogeneous, partially mitigating this concern. Nevertheless, we used filter normalization for 2D surface visualization and report multiple complementary sharpness measures to ensure robustness.

Sharpness and generalization. Foret et al. [4] connected sharpness to generalization via PAC-Bayes bounds. Our CoT models have both lower sharpness and lower validation loss, consistent with this theory. An interesting direction is whether training direct models with SAM would close the generalization gap by explicitly seeking flatter regions, testing whether the landscape geometry is a *cause* or merely a *correlate* of the performance difference.

Mode connectivity. Ainsworth et al. [1] showed that independently trained image classifiers lie in a single basin after permutation alignment. Our finding that CoT models have *higher* same-type barriers (without permutation alignment) suggests either that the single-basin phenomenon does not extend to autoregressive sequence models, or that permutation alignment is essential for revealing the underlying connectivity. Applying Git Re-Basin [1] to transformer-based theorem provers is an important direction for future work.

Neural theorem proving. The step-by-step approach of GPT-f [10] and HTPS [8] provides dense per-step supervision analogous to our CoT format, while Baldur [3] represents the direct approach. Our landscape analysis provides a geometric explanation for the empirical observation that step-by-step methods tend to converge more reliably. DeepSeek-Prover-V2 [11] introduces consistency rewards that explicitly couple CoT structure to formal proofs; this can be interpreted as a landscape-shaping mechanism that further enhances the spectral orthogonality identified in Theorem 3.1.

4.3. **Limitations.** Several limitations constrain the scope of our conclusions.

Scale. Our models have 106,027 parameters, while state-of-the-art theorem provers have billions. Loss landscape geometry may change qualitatively at larger scale. In particular, the spectral orthogonality assumption in Theorem 3.1(ii) may break down in overparameterized regimes where per-step Hessians share significant spectral overlap.

Task complexity. Propositional logic is decidable and admits short proofs. Real theorem proving in dependent type theories (Lean 4, Isabelle) involves complex term construction and genuine mathematical reasoning. Our results establish the existence of geometric differences in a controlled setting but do not guarantee they persist in more complex domains.

Statistical power. Three runs per condition provide limited statistical power. The consistent directional effects are encouraging, but the magnitude of the differences and their statistical significance would benefit from replication with 10 or more seeds.

Confounding factors. CoT outputs are longer than direct outputs. The models therefore process different amounts of target sequence, confounding the generation strategy effect with a sequence length effect. A more careful controlled experiment might fix the total output length and compare the effect of decomposition structure alone.

Reparametrization. While filter normalization addresses some concerns raised by Dinh et al. [2], the Hessian eigenvalue and SAM sharpness measures are not fully reparametrization-invariant. Developing invariant measures suitable for autoregressive models remains an open problem.

5. Conclusion

We have presented the first geometric analysis of loss landscapes in neural theorem proving, comparing chain-of-thought decomposition with direct proof generation. Our main theoretical contribution, Theorem 3.1, shows that the additive structure of the decomposed loss bounds its Hessian spectral norm by the maximum per-step spectral norm when per-step curvature directions are orthogonal. Our experiments on propositional logic theorem proving with matched transformer models confirm that CoT models converge to flatter minima with lower Hessian top eigenvalue, lower SAM sharpness, and smaller gradient norms.

The most notable finding is the flat-but-isolated phenomenon: CoT minima are individually wider but more separated from each other in parameter space, as explained by the barrier superadditivity of Proposition 3.4. This challenges the common assumption that local flatness implies global connectivity and suggests that these are independent geometric properties of the loss landscape.

Several directions merit further investigation.

(1) **Scale.** Does the spectral orthogonality of per-step Hessians persist at the scale of modern theorem provers (10^9+ parameters)?
(2) **Permutation alignment.** Applying Git Re-Basin [1] before barrier measurement may reveal hidden connectivity between CoT solutions.
(3) **SAM intervention.** Training direct models with sharpness-aware minimization [4] would test whether flattening the landscape closes the performance gap.

(4) **Decomposition depth.** Is there a monotone relationship between the number of CoT steps K and the Hessian top eigenvalue? Our theory predicts diminishing returns once the per-step Hessians are approximately orthogonal, but this has not been empirically verified.

(5) **Topological analysis.** Persistent homology of loss sublevel sets could characterize landscape topology beyond the barrier-based measures studied here.

Conjecture 5.1. *There exists a critical model capacity d^* below which CoT and direct models converge to geometrically distinct regions of parameter space, and above which the landscape simplifies to a single basin (modulo permutation symmetries) for both strategies. The critical capacity d^* depends on the complexity of the theorem proving domain.*

Understanding how proof decomposition shapes the optimization landscape is a step toward principled design of neural reasoning systems. The geometric tools developed here—spectral decomposition of the Hessian across reasoning steps, barrier analysis for structured losses, and scale-dependent sharpness comparison—are applicable beyond theorem proving to any setting where the output can be decomposed into intermediate reasoning steps.

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